Kunal Kumar

Embedded Firmware Engineer

RESUME_1296

Location: Chandigarh, Punjab, India | Email: kunal.kumar.resume_1296@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 5 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 5 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: UART, FreeRTOS, Oscilloscope, JTAG, STM32, Embedded Linux, Firmware Debugging, C++, SPI, CAN, ARM Cortex-M, C

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | Meridian Vector Labs | Jan 2023 - Present

- Improved battery runtime by 28% with peripheral and sleep-state optimization
- Validated 443 hardware test vectors across protocol and timing corner cases
- Reduced embedded task jitter by 22% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | OrionPath Technologies | Jul 2021 - Dec 2022

- Reduced embedded task jitter by 12% using interrupt priority and queue tuning
- Lowered field-reset incidents by 35% with watchdog recovery and diagnostics

PROJECTS

Low-Power Metering Device | SPI, C, Oscilloscope, C++

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | FreeRTOS, Embedded Linux, Oscilloscope, Firmware Debugging

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Blue River College of Technology, 2021
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2022)